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Contents

1 Metric Spaces **1**

1.1 Topological Spaces . . . 1

1.1.1 Continuity . . . 2

1.1.2 Sequences . . . 2

1.1.3 Separability . . . 3

1.1.4 Hausdorff topology . . . 3

1.1.5 Metric Topology . . . 3

1.2 Complete Metric Spaces . . . 6

1.2.1 Cauchy Sequences . . . 6

1.2.2 Completion of Metric Spaces . . . 7

1.3 Compactness . . . 10

1.3.1 Totally Bounded Sets . . . 11

1.3.2 Classification of compact sets . . . 12

1.4 Arzelà-Ascoli . . . 13

1.5 Baire Category . . . 16

2 Normed Spaces **19**

2.1 Norms . . . 19

2.2 Finite Dimensional Vector Spaces . . . 20

2.3 $\ell^p(\mathbb{R})$ Spaces . . . 23

2.4 Bounded Linear Transformations . . . 26

2.4.1 The Operator Norm . . . 28

2.4.2 Dual Space . . . 29

Chapter 1

Metric Spaces

1.1 Topological Spaces

Definition 1.1. Given a set X, a topology τ is a subset of the power set $\mathcal{P}(X)$ such that:

i) $\emptyset, X \in \tau$.

ii) τ is closed under union, i.e., if $E_\alpha \in \tau$ for $\alpha \in I$ then $\bigcup_{\alpha \in I} E_\alpha \in \tau$.

iii) τ is closed under finite intersection, i.e., $E_1, E_2, \ldots E_n \in \tau \implies \bigcap_{i=1}^{n} E_i \in \tau$.

Example 1.1. Some examples of topologies are as follows:

- $\tau_{disc} = \mathcal{P}(X)$.
- $\tau_{trivial} = \{\emptyset, X\}$.

Definition 1.2. Let (X, τ) be a topological space with $A \subseteq X$. We have:

1. A is open if and only if $A \in \tau$.
2. A is closed if and only if $A^c \in \tau$.

Definition 1.3. Let (X, τ) be a topological space. A neighborhood of a point $x \in X$ is an open set containing x.

Definition 1.4. Let (X, τ) be a topological space and $A \subseteq X$. We say that x is an interior point of A if and only if there exists an open neighborhood U_x of x contained in A. We denote the set of interior points of A by A°.

Definition 1.5. Let (X, τ) be a topological space. We say that a point $x \in X$ is a limit point of A if and only if for every open neighborhood U_x of x we have

$$U_x \setminus \{x\} \cap A \neq \emptyset.$$

We denote the set of limit points of A by A'.

Definition 1.6. Let (X, τ) be a topological space with $A \subseteq X$. The closure of A is defined as the set $\overline{A} = A \cup A'$.

We have the following properties.

Proposition 1.1. *1. A° is open; in fact, it is the largest open set contained in A, so A is open if and only if $A^\circ = A$.*

2. $\overline{A}$ is closed; in fact, it is the smallest closed set containing A, so A is closed if and only if $A = \overline{A}$ if and only if $A' \subseteq A$.

Proof of 1. If $x \in A^\circ$, then there exists $U_x \subseteq A$. It is clear that $A^\circ \subseteq \bigcup_{x\in A^\circ} U_x$. For the other inclusion, notice that if $y \in U_x$, then U_x is a neighborhood of y and is contained in A, implying that $y \in A^\circ$ and so $U_x \subseteq A^\circ$ for every x. We deduce that $A^\circ = \cup_{x\in A^\circ} U_x$ and so A° is open, being a union of open sets.

Assume G is an open set contained in A. For $x \in G$, there exists U_x such that $U_x \subseteq G \subseteq A^\circ$, and so $x \in A^\circ$ which gives $G \subseteq A^\circ$. Thus, A° is the largest open set contained in A.

Finally, by definition, we have $A^\circ \subseteq A$. If A is open, then by the above, $A \subseteq A^\circ$. Hence $A = A^\circ$.

The proof of 2 is left as an exercise. □

Remark 1.1. As a consequence of the above, A is open if and only if for every $x \in A$ there exists a neighborhood U_x of x such that $U_x \subseteq A$.

1.1.1 Continuity

Definition 1.7. Let (X, τ_X) and (Y, τ_Y) be two topological spaces, and let $f : X \to Y$ be a function, and let $x_0 \in X$. f is continuous at x_0 if and only if for every open neighborhood V of $f(x_0)$ in Y there exists an open neighborhood U of x_0 in X such that $f(U) \subseteq V$.

We say that f is continuous on X if it is continuous at every point of X.

Proposition 1.2. *Let $f : X \to Y$ be a function between two topological spaces. Then f is continuous if and only if $f^{-1}(G)$ is open in X for every open set G in Y.*

1.1.2 Sequences

Definition 1.8. Let (X, τ) be a topological space, and let x_n be a sequence in X. We say that x_n converges to $x_0 \in X$ if and only if for every neighborhood U_{x_0} of x_0 there exists $N \in \mathbb{N}$ such that

$$x_n \in U_{x_0} \quad \forall n \geq N.$$

Definition 1.9. Take the sequence $x_n = \frac{1}{n}$.

- In $(\mathbb{R}, \tau_{disc})$ it does not converge.
- In $(\mathbb{R}, \tau_{trivial})$ it converges to every point in $\mathbb{R}$.

Proposition 1.3. *Let (X, τ) be a topological space, and let x_n be a sequence in A converging to x_0. Then $x_0 \in \overline{A}$.*

Proof. Suppose that $x_0 \notin \overline{A}$. Then there exists a neighborhood U_{x_0} of x_0 such that $U_{x_0} \subseteq (\overline{A})^c = A^c \cap (A')^c$. Since x_n is a sequence in A converging to x_0, $x_n \in U_{x_0}$ for $n \geq N$ for some N, a contradiction since $x_n \in A$. □

Proposition 1.4. *Let $f : (X, \tau_X) \to (Y, \tau_Y)$ be a function between two topological spaces, and let $x_0 \in X$, with f continuous at x_0. If x_n converges to x_0, then $f(x_n)$ converges to $f(x_0)$.*

Remark 1.2. The converse of both propositions above is not true in general.

1.1.3 Separability

Definition 1.10. Let (X, τ) be a topological space and let $A \subseteq X$. We say that A is dense in X if and only if $\overline{A} = X$.

Definition 1.11. Let (X, τ) be a topological space. We say that X is separable if and only if X has a countable dense subset.

Example 1.2. Some familiar examples are:

- $(\mathbb{R}, |\cdot|)$ is separable since $\mathbb{Q}$ is dense in $\mathbb{R}$.
- If X is uncountable, (X, τ_{disc}) is not separable since $\overline{A} = A$ for every $A \subseteq X$ so the only dense subset of X is X itself.

1.1.4 Hausdorff topology

Definition 1.12. Let (X, τ) be a topological space. We say that X is Hausdorff if and only if for every pair of distinct points $x, y \in X$, there exist an open neighborhood U_x of x and an open neighborhood U_y of y such that

$$U_x \cap U_y = \emptyset$$

.

Proposition 1.5. *The limit of a convergent sequence in a Hausdorff space is unique.*

Proof. Assume that (x_n) converges to x and y with $x \neq y$. Since (X, τ) is Hausdorff, there exist neighborhoods U_x of x and U_y of y such that

$$U_x \cap U_y = \emptyset.$$

Since (x_n) converges to x, there exists $N_1 \in \mathbb{N}$ such that $x_n \in U_x$ for all $n \geq N_1$. In particular, this implies that $x_n \notin U_y$ for all $n \geq N_1$, contradicting the fact that (x_n) also converges to y. □

1.1.5 Metric Topology

Definition 1.13. Given a set X, a metric d on X is a function

$$d : X \times X \to \mathbb{R}$$

satisfying the following

i) $d(x, y) \geq 0$ for all $x, y \in X$ with equality if and only if $x = y$.

ii) $d(x, y) = d(y, x)$ for all $x, y \in X$.

iii) $d(x,z) \leq d(x,y) + d(y,z)$ for all $x, y, z \in X$.

We say that (X,d) is a metric space.

Definition 1.14. Given a metric space (X,d), for $x \in X$ and $r > 0$, we define the ball of radius r centered at x as follows

$$B(x,r) = \{y \in X : d(x,y) < r\}.$$

We define $\tau_d = \{U \in \tau_d : \forall x \in U \, \exists r > 0 \, s.t. \, B(x,r) \subseteq U\}$. It is easy to check that τ_d is a Hausdorff topology on X; in fact, it is the smallest topology that contains all balls.

We say that τ_d is the metric topology induced by d.

Definition 1.15. Given a metric space (X,d), we say that $E \subseteq X$ is bounded if and only if $E \subseteq B(x_0, R)$ for some $x_0 \in X$ and $R > 0$. In particular, a sequence is bounded if and only if there exists $x_0 \in X$ and $R > 0$ such that $d(x_n, x_0) < R$ for all $n \in \mathbb{N}$.

Proposition 1.6. *Let (X,d) be a metric space. A sequence x_n converges to x_0 if and only if $\forall \varepsilon > 0 \quad \exists N \in \mathbb{N}$ such that $d(x_n, x_0) < \varepsilon \quad \forall n \geq N$.*

Proposition 1.7. *Let (X, d_X) and (Y, d_Y) be two metric spaces, and let $f : X \to Y$ with $x_0 \in X$. f is continuous at x_0 if and only if $\forall \varepsilon > 0 \quad \exists \delta > 0$ such that $d_Y(f(x), f(x_0)) < \varepsilon$ whenever $d_X(x, x_0) < \delta$ if and only if for every sequence x_n converging to x_0 we have $f(x_n)$ converges to $f(x_0)$.*

Proposition 1.8. *Let (X,d) be a metric space and $E \subseteq X$. Then $x \in \overline{E}$ if and only if there exists a sequence $x_n \in E$ such that x_n converges to x.*

Proof. ($\Longleftarrow$) This is true for any topological space.
($\Longrightarrow$) Let $x \in \overline{E} = E \cup E'$. Then either $x \in E$ or $x \in E'$. If $x \in E$, we may take the constant sequence $x_n = x$ for all n. If $x \in E'$, then by definition of a limit point, for every $n \in \mathbb{N}$ there exists $x_n \in B(x, \frac{1}{n}) \setminus \{x\} \cap E$. Hence $d(x_n, x) < \frac{1}{n}$, so $x_n \to x$ by the Archimedean property. □

We then have the following corollary.

Corollary 1.1. *Given a metric space (X,d) with $E \subseteq X$, E is closed if and only if for every convergent sequence x_n with $x_n \in E$ we have $\lim_{\to\infty} x_n \in E$*

Example 1.3. Some examples of separable and nonseparable metric spaces are:

- $(\mathbb{R}^n, |\cdot|)$ is separable since $\mathbb{Q}^n$ is dense in $\mathbb{R}^n$.
- The space $C([a,b])$ of real-valued continuous functions on $[a,b]$ with the sup-metric (uniform convergence) $d_\infty(f,g) = \sup_{x \in [a,b]} |f(x) - g(x)|$. Let $\mathcal{P}(\mathbb{R}) = \{a_k x^k + \cdots + a_1 x + a_0 : a_i \in \mathbb{R}\}$ be the space of real polynomials. By the Weierstrass approximation theorem, for every $f \in C([a,b])$ there exists a sequence of polynomial functions p_n such that $p_n \to f$ uniformly on $[a,b]$, i.e.

  $$\lim_{n \to \infty} d_\infty(p_n, f) = 0.$$

  For $\varepsilon > 0$, choose $p \in \mathcal{P}(\mathbb{R})$ such that

  $$d_\infty(p, f) < \frac{\varepsilon}{2}.$$

Write

$$p(x) = c_k x^k + \cdots + c_1 x + c_0,$$

with $c_i \in \mathbb{R}$. For each coefficient c_i, choose $q_i \in \mathbb{Q}$ such that

$$|c_i - q_i| < \frac{\varepsilon}{2(k+1)M}, \qquad M = \max_{x\in[a,b]} \{1, |x|, \dots, |x|^k\}.$$

Define

$$q(x) = q_k x^k + \cdots + q_1 x + q_0.$$

Then

$$\begin{aligned} |p(x) - q(x)| &\le \sum_{i=0}^{k} |c_i - q_i|\, |x|^i \\ &\le \sum_{i=0}^{k} \frac{\varepsilon}{2(k+1)M}\, |x|^i \\ &\le \frac{\varepsilon}{2(k+1)M}\, (k+1)M \\ &= \frac{\varepsilon}{2}. \end{aligned}$$

Hence $d_\infty(q, f) \le d_\infty(q, p) + d_\infty(p, f) < \varepsilon$.

We conclude that the countable set $\mathcal{P}(\mathbb{Q})$ is dense in $C([a, b])$ so $C([a, b])$ is separable.

- We define the space of bounded sequences $\ell^\infty(\mathbb{R}) = \{(x_1, x_2, \dots) : x_n \in \mathbb{R},\ \sup_n |x_n| < \infty\}$, with the metric

$$d_\infty(x, y) = \sup_{i\in\mathbb{N}} |x_i - y_i|.$$

Then $\ell^\infty(\mathbb{R})$ is not separable. Suppose, for contradiction, that it is separable, and let M be a countable dense subset. Consider

$$A = \{0, 1\}^{\mathbb{N}},$$

the set of all sequences with entries 0 or 1. The set A is uncountable. If $x, y \in A$ and $x \ne y$, then

$$d_\infty(x, y) = 1.$$

For each $a \in A$, choose a point $m_a \in B(a, \frac{1}{2}) \cap M$. Since

$$B(a, \tfrac{1}{2}) \cap B(b, \tfrac{1}{2}) = \varnothing \quad \text{for } a \ne b,$$

the map $a \mapsto m_a$ is injective. But A is uncountable while M is countable, a contradiction.

Proposition 1.9. *If x_n converges, then it is bounded.*

Definition 1.16. We denote by $c(\mathbb{R})$ the space of convergent sequences. From the above proposition, $c(\mathbb{R}) \subseteq \ell^\infty(\mathbb{R})$.

1.2 Complete Metric Spaces

1.2.1 Cauchy Sequences

Definition 1.17. Given a sequence x_n in X, we say that x_n is Cauchy if and only if $\forall \varepsilon > 0\, \exists N \in \mathbb{N}$ such that $d(x_m, x_n) < \varepsilon\ \forall m, n \geq N$.

Proposition 1.10. *In a metric space (X, d), every convergent sequence is Cauchy, and every Cauchy sequence is bounded.*

Proposition 1.11. *Assume x_n is a Cauchy sequence in a metric space (X, d) with a subsequence converging to a point x_0. Then the sequence x_n converges to x_0.*

Proof. Let $\varepsilon > 0$. Then there exists N_1 such that for all $m, n \geq N_1$ we have $d(x_n, x_m) < \frac{\varepsilon}{2}$. Since $x_{n_k} \to x_0$, there exists N_2 such that for all $k \geq N_2$ we have $d(x_{n_k}, x_0) < \frac{\varepsilon}{2}$. Let $K > N_2$ be such that $n_K \geq N_1$, then for $n \geq N_1$

$$d(x_n, x_0) \leq d(x_n, x_{n_K}) + d(x_{n_K}, x_0) < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon.$$

Hence, $x_n \to x_0$. □

Definition 1.18. We say that (X, d) is complete if and only if every Cauchy sequence converges in X.

Example 1.4.

- $(\mathbb{R}, |\cdot|)$ is complete; this follows from Bolzano Weierstrass.
- $(\mathbb{Q}, |\cdot|)$ is not complete. In fact, by the density of the rationals, there exists a sequence $q_n \in \mathbb{Q}$ that converges to $\sqrt{2}$ in $\mathbb{R}$. Then q_n is Cauchy in $\mathbb{R}$ and so it is Cauchy in $\mathbb{Q}$. But if x_n converges to some $q \in \mathbb{Q}$, then it converges to $q \in \mathbb{R}$ and by uniqueness of limits $q = \sqrt{2} \in \mathbb{Q}$, a contradiction.

Example 1.5. $c(\mathbb{R})$ is closed in $\ell^\infty(\mathbb{R})$.

Solution. Let $(x^n)_{n\geq 1}$ be a sequence in $c(\mathbb{R})$ converging to a sequence $x = (x_1, x_2, \dots)$ in $\ell^\infty(\mathbb{R})$. We write $x^n = (x_{n1}, x_{n2}, \cdots)$. Fix $\varepsilon > 0$. Since $x^n \to x$ in ℓ^∞, there exists $N \in \mathbb{N}$ such that

$$d_\infty(x^N, x) = \sup_{i\in\mathbb{N}} |x_i^N - x_i| < \frac{\varepsilon}{3}.$$

Since $x^N \in c$, it is Cauchy; hence there exists $i_0 \in \mathbb{N}$ such that

$$|x_{Ni} - x_{Nj}| < \frac{\varepsilon}{3} \qquad \forall i, j \geq i_0.$$

Then for all $i, j \geq i_0$,

$$|x_i - x_j| \leq |x_i - x_{Ni}| + |x_{Ni} - x_{Nj}| + |x_{Nj} - x_j| < \frac{\varepsilon}{3} + \frac{\varepsilon}{3} + \frac{\varepsilon}{3} = \varepsilon.$$

Thus $x = (x_i)_{i\geq 1}$ is Cauchy in $\mathbb{R}$, and since $\mathbb{R}$ is complete, (x_i) converges. Therefore $x \in c$. We conclude that c is closed in $\ell^\infty(\mathbb{R})$. □

Example 1.6. Some homework problems.

1. Show that $\ell^\infty(\mathbb{R})$ is complete.
2. Show that $C([a,b])$ is complete with respect to the sup norm.
3. Is $P([a,b])$ closed in $C([a,b])$ with respect to the sup norm?

Proposition 1.12. *Let (X,d) be a metric space and let $E \subseteq X$ be complete. Then E is closed in X.*

Proof. Let (x_n) be a sequence in E that converges to $x \in X$. Being convergent, it is Cauchy in E, and so it converges in E by completeness. Hence, by uniqueness of the limit, we have $x \in E$, so E is closed in X. □

Remark 1.3. The converse is not true in general. For example, consider $(\mathbb{Q}, |\cdot|)$. It is closed in itself but is not complete.

Proposition 1.13. *Every closed subset of a complete metric space is complete, i.e., the converse is true if X is complete.*

Proof. Let (X,d) be a complete metric space and let $E \subseteq X$ be closed. Consider a Cauchy sequence (x_n) in E. Then (x_n) is Cauchy in X, so it converges to some $x \in X$ since X is complete. Since E is closed, $x \in E$, so (x_n) converges in E. □

Example 1.7. $c(\mathbb{R})$ is complete since it is closed in $\ell^\infty(\mathbb{R})$ which is complete, by Examples 1.5 and 1.6.

1.2.2 Completion of Metric Spaces

Lemma 1.1. *If (X,d) is a metric space, then*

$$|d(x,y) - d(x',y')| \leq d(x,x') + d(y,y')$$

Proof. By the triangular inequality $d(x,y) \leq d(x,x') + d(x',y') + d(y',y)$, then

$$d(x,y) - d(x',y') \leq d(x,x') + d(y,y').$$

Reversing the notation and using the symmetric property of d

$$d(x',y') - d(x,y) \leq d(x,x') + d(y,y').$$

This concludes the proof of the lemma. □

Corollary 1.2. *Let (X,d) be a metric space. If x_n, y_n are Cauchy sequences, then $d(x_n,y_n)$ converges in $\mathbb{R}$.*

Proof. Let $d_n = d(x_n,y_n)$ be a sequence in $\mathbb{R}$. Then

$$|d_n - d_m| = |d(x_n,y_n) - d(x_m,y_m)| \leq d(x_n,x_m) + d(y_n,y_m).$$

Since x_n, y_n are Cauchy in X, d_n is Cauchy in $\mathbb{R}$, which by completeness implies that d_n converges. □

Lemma 1.2. *Let (X, d) be a metric space and M a dense subset of X such that every Cauchy sequence in M converges in X. Then X is complete.*

Proof. Let (x_n) be a Cauchy sequence in X. Since M is dense, for every $n \in \mathbb{N}$ there exists $y_n \in M$ such that $d(x_n, y_n) \leq \frac{1}{n}$. Using the triangle inequality,

$$d(y_n, y_m) \leq d(y_n, x_n) + d(x_n, x_m) + d(x_m, y_m) < \frac{1}{n} + d(x_n, x_m) + \frac{1}{m}.$$

Given $\varepsilon > 0$, since x_n is Cauchy, there exists N such that $\frac{1}{N} < \varepsilon/3$ and $d(x_n, x_m) < \varepsilon/3$ for $n, m \geq N$, and so $d(y_n, y_m) < \varepsilon$. We conclude that y_n is Cauchy in M, hence it converges to some $y \in X$.

We show that x_n converges to y. Let $\varepsilon > 0$. There exists N' such that $\frac{1}{N'} < \varepsilon/2$ and $d(y_n, y) < \varepsilon/2$ for $n \geq N'$. Hence for such n,

$$d(x_n, y) \leq d(x_n, y_n) + d(y_n, y) < \frac{1}{n} + d(y_n, y) < \varepsilon.$$

We conclude that x_n converges in X to y, and so X is complete. □

Definition 1.19. A map $\varphi : (X, d_X) \to (Y, d_Y)$ is an isometry if and only if

$$d_Y(\varphi(x), \varphi(y)) = d_X(x, y) \qquad \forall\, x, y \in X.$$

Note that every isometry is injective. If, in addition, ϕ is bijective, we then say that (X, d_X) and (Y, d_Y) are isometric.

Theorem 1.1 (Completion Theorem)**.** *Let (X, d) be a metric space. Then there exists a complete metric space $(\hat{X}, \hat{d})$ and a dense subset M of $\hat{X}$ such that M is isometric to X. $\hat{X}$ is unique up to isometry, and we call it the completion of X.*

Proof. Let C_X be the space of Cauchy sequences in X. We define the relation $\sim$ on C_X as follows

$$x_n \sim y_n \iff \lim_{n\to\infty} d(x_n, y_n) = 0$$

This is indeed an equivalence relation:

- Reflexive: $d(x_n, x_n) = 0$ for all n, so $\lim_{n\to\infty} d(x_n, x_n) = 0$.
- Symmetric: If $x_n \sim y_n$ then since $d(x_n, y_n) = d(y_n, x_n)$ for all n, so
$$0 = \lim_{n\to\infty} d(x_n, y_n) = \lim_{n\to\infty} d(y_n, x_n).$$
- Transitive: If $x_n \sim y_n$ and $y_n \sim z_n$ then $\lim_{n\to\infty} d(x_n, y_n) = 0$ and $\lim_{n\to\infty} d(y_n, z_n) = 0$ and by the triangle inequality we have $d(x_n, z_n) \leq d(x_n, y_n) + d(y_n, z_n) \to 0$, so $x_n \sim z_n$.

Let $\hat{X} = C_X/\sim$ be the set of equivalence classes, so

$$\hat{X} = \{[x_n] : x_n \in C_X\}$$

and define

$$\hat{d}([x_n],[y_n]) = \lim_{n\to\infty} d(x_n, y_n).$$

By Corollary 1.2, $\hat{d}$ exists and is well defined. In fact, if $x_n \sim x'_n$ and $y_n \sim y'_n$ then $|d(x_n, y_n) - d(x'_n, y'_n)| \leq d(x_n, x'_n) + d(y_n, y'_n)$. From Corollary 1.2, $d(x_n, y_n)$ and $d(x'_n, y'_n)$ converge, and since the right-hand side of the inequality goes to zero then $\lim_{n\to\infty} d(x_n, y_n) = \lim_{n\to\infty} d(x'_n, y'_n)$.

Now we show that $\hat{d}$ is a metric on $\hat{X}$:

- $\hat{d}([x_n],[y_n]) \geq 0$ with equality if and only if $\lim_{n\to\infty} d(x_n, y_n) = 0$ if and only if $x_n \sim y_n$ if and only if $[x_n] = [y_n]$.
- $\hat{d}([x_n],[y_n]) = \lim_{n\to\infty} d(x_n, y_n) = \lim_{n\to\infty} d(y_n, x_n) = \hat{d}([y_n],[x_n])$
- Since $d(x_n, z_n) \leq d(x_n, y_n) + d(y_n, z_n)$ for every $x_n, y_n, z_n \in C_X$, letting $n \to \infty$

$$\hat{d}([x_n],[z_n]) \leq \hat{d}([x_n],[y_n]) + \hat{d}([y_n],[z_n]).$$

Now we define $\varphi : X \to \hat{X}$ as follows

$$\varphi(x) = [(x, x, x, x, \ldots)].$$

Notice that

$$\hat{d}(\varphi(x), \varphi(y)) = \lim_{n\to\infty} d(x, y) = d(x, y).$$

Hence φ is an isometry.

We define

$$M = \varphi(X) = \{[(x, x, x, \ldots)] : x \in X\}.$$

We prove that $\overline{M} = \hat{X}$. Take $[x_n] \in \hat{X}$ with (x_n) a Cauchy sequence, we shall show that $B([x_n], \varepsilon) \cap M \neq \emptyset$ for every $\varepsilon > 0$.

Given $\varepsilon > 0$, there exists N such that for $n, m \geq \mathbb{N}$

$$d(x_n, x_m) < \frac{\varepsilon}{2}.$$

Consider the constant sequence (y_n) such that $y_n = x_N$ for every n, so

$$d(x_n, y_n) = d(x_n, x_N) < \frac{\varepsilon}{2} \quad \text{for } n \geq N$$

and

$$\hat{d}([x_n],[y_n]) = \lim_{n\to\infty} d(x_n, y_n) \leq \frac{\varepsilon}{2} < \varepsilon.$$

Hence, $[y_n] = \varphi(x_N) = [(x_N, x_N, \ldots)] \in B([x_n], \varepsilon) \cap M$ and hence $\overline{M} = \hat{X}$.

It remains to prove that $\hat{X}$ is complete. We use Lemma 1.2, take a Cauchy sequence p_{x_n} in M as follows

$$\begin{aligned}
p_{x_1} &= \varphi(x_1) = [(x_1, x_1, x_1, \ldots)] \\
p_{x_2} &= \varphi(x_2) = [(x_2, x_2, x_2, \ldots)] \\
&\vdots \\
p_{x_n} &= \varphi(x_n) = [(x_n, x_n, x_n, \ldots)]
\end{aligned}$$

Consider the sequence $(x_n) = (x_1, x_2, x_3, \ldots)$, we have

$$d(x_n, x_m) = \hat{d}(\varphi(x_n), \varphi(x_m)) = \hat{d}(p_{x_n}, p_{x_m}).$$

Since p_{x_n} is Cauchy in $\hat{X}$ we deduce that (x_n) is Cauchy in X, i.e $(x_n) \in C_X$.

Define then $p = [(x_n)] \in \hat{X}$. Let $\varepsilon > 0$, there exists N such that for $n, k \geq N$ we have $d(x_n, x_k) \leq \frac{\varepsilon}{2}$, and so for $k \geq N$

$$\hat{d}(p, p_{x_k}) = \lim_{n\to\infty} d(x_n, x_k) \leq \frac{\varepsilon}{2} < \varepsilon$$

which shows that p_{x_k} converges to p in $\hat{X}$, so by lemma 1.2 $\hat{X}$ is complete.

Uniqueness is proved in the problem set. □

1.3 Compactness

Definition 1.20. Let (X, τ) be a topological space and $K \subseteq X$. We say that K is compact if and only if every open cover $\mathcal{G} = \{G_\alpha\}_{\alpha\in I}$ of K has a finite subcover.

Example 1.8. $[0, 1)$ is not compact since the open cover $\{(-1, 1 - \frac{1}{n})\}_{n\in\mathbb{N}}$ has no finite subcover.

Example 1.9. Consider a finite subset $A = \{a_1, a_2, a_3, \ldots, a_N\}$ of a toplogical space X and let $\mathcal{G} = \{G_\alpha\}_\alpha$ be an open cover of A, i.e. $A \subseteq \bigcup_{\alpha\in I} G_\alpha$. For each i, there exists $\alpha_i \in I$ such that $a_i \in G_{\alpha_i}$ so $\{G_{\alpha_1}, G_{\alpha_2}, \ldots, G_{\alpha_N}\}$ is a finite subcover of $\mathcal{G}$, deducing that A is compact.

Proposition 1.14. *In a topological space (X, τ), every closed subset of a compact set is compact.*

Proof. Consider a compact set K and a closed subset E of K. Let $\mathcal{G} = \{G_\alpha\}_{\alpha\in I}$ be an open cover of E, so $E \subseteq \bigcup_{\alpha\in I} G_\alpha$ then

$$K \subset E^c \cup \bigcup_{\alpha\in I} G_\alpha.$$

Since E is closed, E^c is open so $\mathcal{G} \cup \{E^c\}$ is an open cover of K. Then there exists a finite subcover $\{G_{\alpha_1}, G_{\alpha_2}, \ldots, G_{\alpha_n}, E^c\}$ of K. Therefore,

$$E = E \cap K \subseteq \bigcup_{i=1}^{n} G_{\alpha_i}$$

deducing that $\{G_{\alpha_1}, G_{\alpha_2}, \ldots, G_{\alpha_n}\}$ is a finite subcover of $\mathcal{G}$ so E is compact. □

Proposition 1.15. *If (X, τ) is a Hausdorff space and $K \subseteq X$ is compact then K is closed.*

Proof. We show that K^c is open. Take $q \in K^c$. Since X is Hausdorff, for every $p \in K$ there exist a neighborhood U_p of p and a neighborhood V_p^q of q such that $U_p \cap V_p^q = \emptyset$.

$\mathcal{G} = \{U_p\}_{p\in K}$ is then an open cover of the compact set K, so it has a finite subcover, i.e.

$$K \subseteq U_{p_1} \cup U_{p_2} \cup \ldots \cup U_{p_n}.$$

Let $V^q = V_{p_1}^q \cap V_{p_2}^q \cap \ldots \cap V_{p_n}^q$. V^q is an open neighborhood of q, and

$$V^q \cap K \subseteq V^q \cap (U_{p_1} \cup U_{p_2} \cup \ldots \cup U_{p_n}) = \emptyset.$$

Then $V^q \subseteq K^c$, which shows that K^c is open so K is closed. □

Proposition 1.16. *In a metric space (X, d), if $K \subseteq X$ is compact then K is closed and bounded.*

Example 1.10. Consider again $\ell^\infty(\mathbb{R})$ with d_∞, and $A = \{0,1\}^{\mathbb{N}}$. Clearly A is bounded. In fact, taking $\mathbf{0} = (0, 0, 0, \dots)$, we have $d(x, \mathbf{0}) \leq 1$ for every $x \in A$. A is closed since, if $y = (y_1, y_2, \cdots)$ in A^c there exists $y_i \notin \{0,1\}$ and there exists $\delta > 0$ such that $(y_i - \delta, y_i + \delta) \cap \{0,1\} = \emptyset$. We get that $B(y, \delta) \subseteq A^c$ so A^c is open, which shows that A is closed.

Although A is closed and bounded, A is not compact. In fact, let $\mathcal{G} = \{B(x, \frac{1}{2}) : x \in A\}$. Notice that $B(x, \frac{1}{2}) \cap A = \{x\}$ but since A is infinite, $\mathcal{G}$ has no finite subcover.

1.3.1 Totally Bounded Sets

Definition 1.21. Let (X, d) be a metric space and $K \subseteq X$. We say that K is totally bounded if and only if $\forall \varepsilon > 0$ there exists $x_1, \dots, x_n \in K$ such that $K \subseteq \bigcup_{i=1}^n B(x_i, \varepsilon)$.

Example 1.11. Following the same reasoning in Example 1.10, $A = \{0,1\}^{\mathbb{N}}$ is not totally bounded in $\ell^\infty(\mathbb{R})$.

Remark 1.4. It is easy to check that every compact set K in a metric space X is totally bounded. In fact, given $\varepsilon > 0$, cover K with balls centered at points of K and radius ε. By compactness, the constructed cover has a finite subcover.

Proposition 1.17. *If K is totally bounded, then K is bounded.*

Proof. Let $\varepsilon = 1$. Then there exists $x_1, \dots, x_n \in K$ such that $K \subseteq \bigcup_{i=1}^n B(x_i, 1)$. For every $x \in K$ there exists i such that $d(x, x_i) < 1$ then $d(x, x_1) \leq d(x, x_i) + d(x_i, x_1) < 1 + d(x_i, x_1)$. Let $M = 1 + \max\{d(x_i, x_1) : i = 1, \dots, n\}$ then $d(x, x_1) < M$ for all $x \in K$ so K is bounded. □

Proposition 1.18. *A subset of a totally bounded set is totally bounded.*

Proof. Let K be totally bounded and $E \subseteq K$. Fix $\varepsilon > 0$. Then we have $x_1, x_2, \cdots, x_n \in K$ s.t.

$$K \subseteq B(x_1, \varepsilon/2) \cup \cdots \cup B(x_n, \varepsilon/2)$$

Pick $y_1, \cdots, y_k$ with $y_i \in B(x_i, \varepsilon/2) \cap E$ (skipping the empty ones). Then we have that

$$E \subseteq B(y_1, \varepsilon) \cup \cdots \cup B(y_k, \varepsilon)$$

□

Proposition 1.19. *In $(\mathbb{R}^n, |\cdot|)$, K is totally bounded if and only if K is bounded.*

Proof. The forward direction is proved for general metric spaces. To prove the converse, assume K is bounded. Then there exists a closed cube in $\mathbb{R}^n$ such that $K \subseteq Q$. Q is compact, so it is totally bounded, and hence $K \subseteq Q$ is totally bounded. □

1.3.2 Classification of compact sets

We show the following theorem that classifies compactness in general metric spaces.

Theorem 1.2. *Let (X, d) be a metric space and $K \subseteq X$. Then the following are equivalent:*

1. *K is compact*
2. *Every infinite subset E of K has a limit point in K (Point compactness).*
3. *Every sequence in K has a convergent subsequence with limit in K (sequential compactness).*
4. *K is complete and totally bounded.*

Proof.

- (1) $\Rightarrow$ (2). If $E \subseteq K$ is infinite, then E has no limit point in K. Then for every $x \in K$ there exists an open neighborhood U_x of x such that $U_x \setminus \{x\} \cap E = \emptyset$. Then $\{U_x\}_{x \in K}$ is an open cover of K. Since K is compact, there exists a finite subcover $\{U_{x_1}, \ldots, U_{x_n}\}$ of K. Then
$$E = E \cap K \subseteq \bigcup_{i=1}^{n} E \cap U_{x_i} \subseteq \{x_1, \cdots, x_n\},$$
a contradiction.

- (2) $\Rightarrow$ (3). Let x_n be a sequence in K. Define the set $E = \{x_n, n \in \mathbb{N}\}$. If E is finite, then there exists $n_1 < n_2 < \ldots$ and $x_0 \in E$ such that $x_{n_k} = x_0$ for all k, so $x_{n_k} \to x_0$. If E is infinite, then by (2) there exists a limit point $x_0 \in K$ of E. Let $\varepsilon = 1$, there exists n_1 such that $x_{n_1} \in B(x_0, 1) \cap E \neq \emptyset$. Having $x_{n_{k-1}}$, $B(x_0, \frac{1}{k}) \cap E \neq \emptyset$; in fact, it is infinite. Then there exists $n_k > n_{k-1}$ such that $x_{n_k} \in B(x_0, \frac{1}{k}) \cap E$. The sequence x_{n_k} then converges to x_0.

- (3) $\Rightarrow$ (4).

  Completeness. Let (x_n) be a Cauchy sequence in K. Since K is sequentially compact, (x_n) has a convergent subsequence $x_{n_k} \to x_0 \in K$. Hence, by Proposition 1.11, $x_n \to x_0 \in K$. Hence K is complete.

  Total boundedness. Suppose, for contradiction, that K is not totally bounded. Then there exists $\varepsilon_0 > 0$ such that K cannot be covered by finitely many balls of radius ε_0 centered at points of K.

  We construct a sequence (x_n) in K as follows. Pick $x_1 \in K$ arbitrarily. Then take $x_2 \in K \setminus B(x_1, \varepsilon_0)$ and continue recursively. Having $x_1, \ldots, x_n \in K$,
  $$B(x_1, \varepsilon_0) \cup \cdots \cup B(x_n, \varepsilon_0)$$
  does not cover K, and there exists
  $$x_{n+1} \in K \setminus \bigcup_{i=1}^{n} B(x_i, \varepsilon_0).$$
  Notice that for $m \neq n$, $d(x_m, x_n) \geq \varepsilon_0$, so no subsequence of x_n would converge, a contradiction. Hence K is totally bounded.

- (4) $\Rightarrow$ (1). Assume K is not compact. Then there exists $\mathcal{G} = \{G_\alpha\}_{\alpha\in I}$ an open cover of K with no finite subcover. Since K is totally bounded, K can be covered by finitely many balls of radius 1 centered at points of K. Then, there exists $x_1 \in K$ and $B(x_1, 1)$ such that $K_1 = B(x_1, 1) \cap K$ cannot be covered by finitely many G_α. Now, since $K_1 \subseteq K$, K_1 is totally bounded and, as above, there exist $x_2 \in K_1$ and $B(x_2, \frac{1}{2})$ such that $K_2 = B(x_2, \frac{1}{2}) \cap K_1$ cannot be covered by finitely many G_α.

  Recursively, we construct a sequence of nested sets

  $$K =: K_0 \supseteq K_1 \supseteq K_2 \supseteq \cdots$$

  with $K_n = B(x_n, \frac{1}{n}) \cap K_{n-1}$, $x_n \in K_{n-1}$, and K_n cannot be covered by finitely many G_α.

  We show that x_n is Cauchy. Let $m, n > N$. Then $x_n, x_m \in K_{n-1}, K_{m-1} \subseteq K_N \subseteq B(x_N, 1/N)$. Thus,

  $$d(x_n, x_m) \leq d(X_n, x_N) + d(x_m, x_N) < \frac{2}{N}.$$

  For $\varepsilon > 0$, choose N such that $\frac{2}{N} < \varepsilon$, we get that $d(x_n, x_m) < \varepsilon$ for $n, m > N$ which shows that x_n is Cauchy.

  Since K is complete, x_n converges to $x_0 \in K$. Since $\mathcal{G}$ is an open cover of K, there exists $\alpha_0 \in I$ such that $x_0 \in G_{\alpha_0}$. Since G_{α_0} is open, there exists $\delta > 0$ such that $B(x_0, \delta) \subseteq G_{\alpha_0}$. Choose N such that $\frac{1}{N} < \delta/2$, and $d(x_N, x_0) < \delta/2$. Notice that

  $$K_N \subseteq B(x_N, 1/N) \subseteq B(x_0, \delta) \subseteq G_{\alpha_0}.$$

  A contradiction. Hence K is compact.

□

1.4 Arzelà-Ascoli

Proposition 1.20. *Let $f : (X, \tau_X) \to (Y, \tau_Y)$ be continuous and let $K \subseteq X$ be compact. Then $f(K)$ is compact in Y.*

Proof. Use definition. □

Corollary 1.3. *Given a metric space (X, d), a compact set $K \subseteq X$, and $f : X \mapsto \mathbb{R}$, a continuous function. Then, there exists $p, q \in K$ such that*

$$f(p) = \sup_K f(x), \quad f(q) = \inf_K f(x).$$

Definition 1.22. Let X be a compact metric space. We denote by $C(X)$ the space of real-valued continuous functions on X. Define the metric

$$d_\infty(f, g) = \sup_{x\in X} |f(x) - g(x)|.$$

The space $(C(X), d_\infty)$ is complete (because uniformly Cauchy in $\mathbb{R}$ implies uniformly convergent) and is separable (follows from the general Stone-Weierstrass theorem).

In all this section, we consider $C(X)$ with the d_∞ metric

Remark 1.5. Recall that, if X is compact and $f : X \to Y$ is continuous, then f is uniformly continuous, i.e.

$$\forall \varepsilon > 0 \ \exists \delta > 0 \text{ such that } d_Y(f(x), f(y)) < \varepsilon \quad \text{for all } x, y \in X \text{ with } d_X(x, y) < \delta.$$

Definition 1.23. $\mathcal{F} \subseteq C(X)$ is equicontinuous if and only if for every $\varepsilon > 0$ there exists $\delta > 0$ such that $d_X(x, y) < \delta$ implies $|f(x) - f(y)| < \varepsilon$ for all $f \in \mathcal{F}$.

Theorem 1.3 (Arzelà-Ascoli)**.** *Let X be a compact metric space. $\mathcal{F} \subseteq C(X)$ is totally bounded if and only if $\mathcal{F}$ is equicontinuous and bounded.*

Proof. ($\Longrightarrow$) $\mathcal{F}$ is bounded since it is totally bounded.

Let $\varepsilon > 0$. Since $\mathcal{F}$ is totally bounded, there exists $f_1, \dots f_n \in \mathcal{F}$ such that $\mathcal{F} \subseteq B(f_1, \frac{\varepsilon}{3}) \cup \dots \cup B(f_n, \frac{\varepsilon}{3})$. Now $f_i \in C(X)$, so it is uniformly continuous. Thus, there exists $\delta_i > 0$ such that $|f_i(x) - f_i(y)| < \frac{\varepsilon}{3}$ for all $x, y \in X$ with $d_X(x, y) < \delta_i$. Let $\delta = \min\{\delta_1, \dots, \delta_n\}$. Take $f \in \mathcal{F}$ and $x, y \in X$ such that $d(x, y) < \delta$. Then there exists $i_0 \in \{1, \dots, n\}$ such that $f \in B(f_{i_0}, \frac{\varepsilon}{3})$ and so

$$|f(x)-f(y)| \leq |f(x)-f_{i_0}(x)|+|f_{i_0}(x)-f_{i_0}(y)|+|f_{i_0}(y)-f(y)| \leq 2d_\infty(f, f_{i_0})+|f_{i_0}(x)-f_{i_0}(y)| < \varepsilon$$

Therefore, $\mathcal{F}$ is equicontinuous.
($\Longleftarrow$) Let $\varepsilon > 0$. Then there exists δ such that $\forall f \in \mathcal{F}$ we have

$$|f(x) - f(y)| < \frac{\varepsilon}{3} \quad \forall x, y \in X \quad \text{such that} \quad d(x, y) < \delta.$$

Since X is compact, we can cover X with finitely many balls of radius δ, i.e.,

$$X = B(x_1, \delta) \cup B(x_2, \delta) \cup \dots \cup B(x_N, \delta).$$

Define the map

$$\varphi : \mathcal{F} \to (\mathbb{R}^N, |\cdot|)$$

as follows:

$$\varphi(f) = (f(x_1), \dots, f(x_N)).$$

We claim that $\varphi(\mathcal{F})$ is bounded in $\mathbb{R}^N$. In fact, since $\mathcal{F}$ is bounded in $C(X)$, there exists $M > 0$ such that $|f(x)| \leq M$ for all $f \in \mathcal{F}$ and $x \in X$. Then

$$|\varphi(f)| = \sqrt{\sum_{i=1}^{N} |f(x_i)|^2} \leq \sqrt{N} M,$$

and the claim follows.

Therefore, $\varphi(\mathcal{F})$ is totally bounded in $\mathbb{R}^n$ (or simply $\overline{\varphi(\mathcal{F})}$ is compact). Then there exists $f_1, \dots, f_K \in \mathcal{F}$ such that

$$\varphi(\mathcal{F}) \subseteq B(\varphi(f_1), \frac{\varepsilon}{3}) \cup \dots \cup B(\varphi(f_K), \frac{\varepsilon}{3}).$$

We prove that $\mathcal{F} \subseteq B(f_1, \varepsilon) \cup \dots \cup B(f_K, \varepsilon)$. Let $f \in \mathcal{F}$. Then there exists $i_0 \in \{1, \dots, K\}$ such that $\varphi(f) \in B(\varphi(f_{i_0}), \frac{\varepsilon}{3})$ and so for every $j = 1, \cdots, N$

$$|f(x_j) - f_{i_0}(x_j)| \leq |\varphi(f) - \varphi(f_{i_0})| < \varepsilon/3.$$

Let $x \in X$. Then $x \in B(x_{j_0}, \delta)$ for some $j_0 \in \{1, \dots, N\}$. Then

$$|f(x) - f_{i_0}(x)| \leq |f(x) - f(x_{j_0})| + |f(x_{j_0}) - f_{i_0}(x_{j_0})| + |f_{i_0}(x_{j_0}) - f_{i_0}(x)| < \frac{\varepsilon}{3} + \frac{\varepsilon}{3} + \frac{\varepsilon}{3} = \varepsilon.$$

Hence,

$$d_\infty(f, f_{i_0}) = \sup_X |f(x) - f_{i_0}(x)| = \max_X |f(x) - f_{i_0}(x)| < \varepsilon$$

so $f \in B(f_{i_0}, \varepsilon)$ and therefore $\mathcal{F} \subseteq B(f_1, \varepsilon) \cup \ldots \cup B(f_K, \varepsilon)$ which shows that $\mathcal{F}$ is totally bounded. □

Since closed subsets of complete sets are complete, and compactness is equivalent to completeness and total boundedness, we get the following result.

Corollary 1.4. *$\mathcal{F} \subseteq C(X)$ is compact if and only if $\mathcal{F}$ is equicontinuous, bounded and closed.*

Lemma 1.3. *If $\mathcal{F} \subseteq C(X)$ is equicontinuous, then $\overline{\mathcal{F}}$ is equicontinuous.*

Proof. Let $\varepsilon > 0$. Then there exists $\delta > 0$ such that $d_X(x, y) < \delta$ implies $|f(x) - f(y)| < \varepsilon/2$ for all $f \in \mathcal{F}$.

Let $g \in \overline{\mathcal{F}}$. There exists a sequence $g_n \in \mathcal{F}$ such that $g_n \to g$ uniformly. We have for $x, y \in X$ with $d(x, y) < \delta$

$$|g_n(x) - g_n(y)| < \varepsilon/2.$$

Letting $n \to \infty$, we get

$$|g(x) - g(y)| \leq \varepsilon/2 < \varepsilon,$$

for every $g \in \mathcal{F}$, which shows that $\overline{\mathcal{F}}$ is equicontinuous. □

Theorem 1.4 (Sequential Arzelà-Ascoli)**.** *Let f_n be a sequence in $C(X)$ such that*

i) f_n is equicontinuous, i.e., for every $\varepsilon > 0$ there exists $\delta > 0$ such that for all $n \in \mathbb{N}$, $|f_n(x) - f_n(y)| < \varepsilon$ for $d(x, y) < \delta$.

ii) f_n is uniformly bounded, i.e., there exists $M > 0$ such that $|f_n(x)| \leq M$ for all $n \in \mathbb{N}$ and all $x \in X$.

Then there exists a subsequence f_{n_k} that converges uniformly to some $f \in C(X)$.

Proof. $\mathcal{F} = \{f_n : n \in \mathbb{N}\}$ is equicontinuous and bounded. Then, from Corollary 1.4, $\overline{\mathcal{F}}$ is closed, bounded and equicontinuous and, by Theorem 1.3 $\overline{\mathcal{F}}$ is compact, hence sequentially compact. Therefore, there exist a subsequence $f_{n_k} \to f$ uniformly for some $f \in \overline{\mathcal{F}}$. □

Example 1.12. For $\alpha \in \mathbb{R}$, define the set $\mathcal{F}_\alpha = \{f \in Lip_1([0, 1]) : f(0) = \alpha\}$ where $Lip_1([0, 1]) = \{f : |f(x) - f(y)| \leq |x - y|, x, y \in [0, 1]\}$. Show that $\mathcal{F}_\alpha$ is compact in $C([0, 1])$, then deduce that if $g \in C([0, 1])$ then there exists a function $f \in Lip_1([0, 1])$ such that $f(0) = g(0)$

$$d_\infty(f, g) = \inf\{d_\infty(h, g) : h \in Lip_1([0, 1]), h(0) = g(0)\}.$$

Solution. Notice that $\mathcal{F}_\alpha$ is closed. In fact, if $f_n \in \mathcal{F}_\alpha$ converges to f, then

$$f(0) = \lim_{n\to\infty} f_n(0) = \alpha.$$

Also

$$|f(x) - f(y)| = \lim_{n\to\infty} |f_n(x) - f_n(y)| \leq |x - y|.$$

Thus, $f \in \mathcal{F}_\alpha$.

Next we prove that $\mathcal{F}_\alpha$ is bounded. In fact, for $f \in \mathcal{F}_\alpha$

$$|f(x)| \leq |f(x) - f(0)| + |f(0)| \leq |x| + |\alpha| \leq 1 + |\alpha|.$$

Finally, $\mathcal{F}_\alpha$ is equicontinuous by using the Lipschitz condition and taking $\delta = \varepsilon$. We conclude from Corollary 1.4 that $\mathcal{F}_\alpha$ is compact.

For the second part, apply sequential compactness to $\mathcal{F}_\alpha$ with $\alpha = g(0)$.

1.5 Baire Category

Definition 1.24. Let (X, τ) be a topological space and $A \subseteq X$. We say that A is nowhere dense if and only if $(\overline{A})^\circ = \emptyset$.

Example 1.13. Some examples are:

- The singletons $\{x\}$ in $(\mathbb{R}, |\cdot|)$ are nowhere dense since $\overline{\{x\}} = \{x\}$ and $\{x\}^\circ = \emptyset$.
- Closed sets with empty interiors are nowhere dense; for example, $\mathbb{N}$ and $\mathbb{Z}$ are nowhere dense in $(\mathbb{R}, |\cdot|)$.
- $\mathbb{Q}$ is not nowhere dense in $\mathbb{R}$ since $\overline{\mathbb{Q}} = \mathbb{R}$ and $\mathbb{R}^\circ = \mathbb{R} \neq \emptyset$.

Proposition 1.21. *A is nowhere dense in X if and only if $X \setminus \overline{A}$ is dense in X.*

Proof. ($\Longrightarrow$) Let $x \in X$ and U_x be an open neighborhood of x. Since A is nowhere dense, $U_x \cap X \setminus \overline{A} \neq \emptyset$ which shows that $\overline{X \setminus \overline{A}} = X$.
($\Longleftarrow$) Similar to above. □

Definition 1.25. Let (X, τ) be a topological space and $A \subseteq X$. We say that A is first category or meagre if and only if A can be written as a countable union of nowhere dense subsets of X. Otherwise, we say that A is second category or non-meagre.

Example 1.14. For example, $\mathbb{Q}$ is first category in $\mathbb{R}$ since $\mathbb{Q} = \bigcup_{q\in\mathbb{Q}} \{q\}$ and $\{q\}$ is nowhere dense in $\mathbb{R}$.

Theorem 1.5 (Baire Category Theorem)**.** *Let (X, d) be a complete metric space. If $G_1, G_2, \ldots$ are open dense subsets of X, then $\bigcap_{n=1}^\infty G_n$ is dense in X. Equivalently, every non-empty open subset of X is second category.*

In particular, if $X = \bigcup_{n=1}^\infty F_n$ with F_n closed then there exists n_0 such that F_{n_0} has non-empty interior.

Question 1.1 (Why equivalently?)**.** ($\Longrightarrow$) Let G be open in X and suppose

$$G = \bigcup_{n=1}^{\infty} A_n \subseteq \bigcup_{n=1}^{\infty} \overline{A_n}$$

with A_n nowhere dense. Then $G_n = X \setminus \overline{A_n}$ is open and dense. Thus,

$$\bigcap_{n=1}^{\infty} X \setminus \overline{A_n} = \left(\bigcup_{n=1}^{\infty} \overline{A_n} \right)^c$$

is dense in X. Contradiction, since $G \cap \left(\bigcup_{n=1}^{\infty} \overline{A_n} \right)^c = \emptyset$.
($\Longleftarrow$) Let $G_1, G_2, \ldots$ be open dense subsets of X; each G_i second category. Suppose that $\bigcap_{n=1}^{\infty} G_n$ is not dense. Then there exists non-empty open set H such that

$$H \subseteq \left(\bigcap_{n=1}^{\infty} G_n \right)^c = \bigcup_{n=1}^{\infty} G_n^c$$

Thus,

$$H = \bigcup_{n=1}^{\infty} G_n^c \cap H.$$

Being open, H is second category, there exists n_0 such that $G_{n_0}^c \cap H$ is not nowhere dense, and so

$$\overline{G_{n_0}^c \cap H} \subseteq \overline{G_{n_0}^c} = G_{n_0}^c$$

has non-empty interior, a contradiction, since G_{n_0} is dense in X.

Proof of BCT. Let $x_0 \in X$ and $\delta > 0$, and set $B_0 = B(x_0, \delta)$. We shall show that

$$B(x_0, \delta) \cap (\cap_{n=1}^{\infty} G_n) \neq \emptyset.$$

Since G_1 is dense in X, $B_0 \cap G_1 \neq \emptyset$. Take $x_1 \in B_0 \cap G_1$. Since G_1 is open, choose $\delta_1 < \frac{\delta}{2}$, such that

$$B_1 = B(x_1, \delta_1) \subseteq \overline{B(x_1, \delta_1)} \subseteq B_0 \cap G_1.$$

Since G_2 is dense in X, $B_1 \cap G_2 \neq \emptyset$. Thus, there exists $x_2 \in B_1 \cap G_2$ and $\delta_2 \leq \frac{\delta}{2^2}$ such that

$$B_2 = B(x_2, \delta_2) \subseteq \overline{B(x_2, \delta_2)} \subseteq B_1 \cap G_2 \subseteq B_0 \cap G_1 \cap G_2.$$

Proceeding inductively, suppose we have x_{n-1}, $\delta_{n-1} < \frac{\delta}{2^{n-1}}$, and $B_{n-1} = B(x_{n-1}, \delta_{n-1})$, since G_n is dense in X, $B_{n-1} \cap G_n \neq \emptyset$. Therefore, there exist $x_n \in B_{n-1} \cap G_n$ and $\delta_n < \frac{\delta}{2^n}$ such that,

$$B_n = B(x_n, \delta_n) \subseteq \overline{B(x_n, \delta_n)} \subseteq B_{n-1} \cap G_n.$$

Consequently,

$$B_n \subseteq B_0 \cap G_1 \cap \cdots \cap G_n.$$

We prove that (x_n) is Cauchy. Let $n, m > N$. Since the balls are nested,

$$x_n, x_m \in B_N = B(x_N, \delta_N).$$

Therefore, by the triangle inequality,

$$d(x_n, x_m) \leq d(x_n, x_N) + d(x_m, x_N) < 2\delta_N < \frac{\delta}{2^{N-1}}.$$

Given $\varepsilon > 0$, choose $N \in \mathbb{N}$ sufficiently large such that $\frac{\delta}{2^{N-1}} < \varepsilon$. Then, for all $n, m > N$,

$$d(x_n, x_m) < \varepsilon.$$

Hence, (x_n) is a Cauchy sequence.

Since X is complete, x_n converges to an element $z \in X$. Fix m. For all $n > m$, we have $x_n \in \overline{B_m}$, and since $\overline{B_m}$ is closed

$$z \in \overline{B_m} \subseteq B_0 \cap G_1 \cap \cdots \cap G_m \qquad \forall m.$$

This implies

$$z \in B_0 \cap \left(\bigcap_{n=1}^{\infty} G_n \right).$$

We have shown that for every $x_0 \in X$ and $\delta > 0$ $B(x_0, \delta) \cap \bigcap_{n=1}^{\infty} G_n \neq \emptyset$, so $\bigcap_{n=1}^{\infty} G_n$ is dense in X. □

Chapter 2

Normed Spaces

2.1 Norms

Definition 2.1. Given a vector space V over $\mathbb{F} = \mathbb{R}$ or $\mathbb{C}$, a norm is a map $\|\cdot\| : V \to \mathbb{R}$ satisfying

i) $\|x\| \geq 0$, with equality if and only if $x = 0$.

ii) $\|\alpha x\| = |\alpha| \cdot \|x\|$, $\alpha \in \mathbb{F}$, $x \in V$.

iii) $\|x + y\| \leq \|x\| + \|y\|$, $\forall x, y \in V$.

Example 2.1. $\ell^\infty(\mathbb{R})$ (the set of bounded sequences) under coordinate-wise addition and scalar multiplication and

$$\|x\|_\infty = \sup_{i \in \mathbb{N}} |x_i| = d_\infty(x, 0).$$

$C(X)$ with X compact and

$$\|f\|_\infty = \sup_{x \in X} |f(x)|.$$

Proposition 2.1. *Every norm induces a metric*

$$d(x, y) = \|x - y\|.$$

This allows us to deal with normed spaces as metric spaces with respect to the induced metric.

Remark 2.1. Given a norm $\|\cdot\|$ on V, we will denote by $(V, \|\cdot\|)$ the metric space V with the corresponding metric induced by $\|\cdot\|$. Notice that, by continuity of the metric, the norm is a continuous map from V to $\mathbb{R}$.

Definition 2.2. Given a vector space V and two norms $\|\cdot\|_a, \|\cdot\|_b$ on V, we say that $\|\cdot\|_a$ and $\|\cdot\|_b$ are equivalent if and only if they induce the same topology.

Proposition 2.2. *Given a vector space V and two norms $\|\cdot\|_a, \|\cdot\|_b$ on V, the following are equivalent:*

i) $\|\cdot\|_a$ and $\|\cdot\|_b$ are equivalent.

ii) $\exists C_1, C_2 > 0$ such that

$$C_2 \|x\|_b \leq \|x\|_a \leq C_1 \|x\|_b, \quad \forall x \in V.$$

iii) $x_n \to x_0$ *in* $(V, \|\cdot\|_a) \iff x_n \to x_0$ *in* $(V, \|\cdot\|_b)$.

Proof. [*i)* $\implies$ *ii)*] $B_{\|\cdot\|_a}(0,1) = \{x \in V,\ \|x\|_a < 1\}$ is open in $(V, \|\cdot\|_a)$, and hence $B_{\|\cdot\|_a}(0,1)$ is open in $(V, \|\cdot\|_b)$. Therefore, 0 is an interior point of $B_{\|\cdot\|_a}(0,1)$ with respect to the $\|\cdot\|_b$ norm, and so there exists $\delta > 0$ such that

$$B_{\|\cdot\|_b}(0,\delta) \subseteq B_{\|\cdot\|_a}(0,1).$$

Now take $x \in V$, $x \neq 0$ (the inequality holds trivially at 0). Then

$$\frac{\delta}{2}\frac{x}{\|x\|_b} \in B_{\|\cdot\|_b}(0,\delta) \subseteq B_{\|\cdot\|_a}(0,1).$$

Hence

$$\left\|\frac{\delta}{2}\frac{x}{\|x\|_b}\right\|_a < 1 \implies \|x\|_a < \frac{2}{\delta}\|x\|_b .$$

Switching $\|\cdot\|_a$ and $\|\cdot\|_b$, we similarly obtain the other inequality.

[*ii)* $\implies$ *iii)*] This follows immediately from the squeeze theorem and the following inequality:

$$C_2 \|x_n - x_0\|_b \leq \|x_n - x_0\|_a \leq C_1 \|x_n - x_0\|_b .$$

[*iii)* $\implies$ *i)*] A set is closed in $(V, \|\cdot\|_a)$ if and only if it is closed in $(V, \|\cdot\|_b)$, by the sequential property of closed sets.

□

Definition 2.3. A complete normed space $(V, \|\cdot\|)$ is called a Banach space.

Example 2.2. The space of polynomial $\mathcal{P}([a,b])$ is not Banach in the sup norm, whereas ℓ^1 and ℓ^∞ are Banach.

2.2 Finite Dimensional Vector Spaces

Definition 2.4. Let V be a finite-dimensional vector space and $S = \{v_1, \cdots, v_n\}$ a basis of V. Take $x \in V$. Then there exists unite $r_1, r_2, \cdots, r_n$ such that

$$x = r_1 v_1 + \cdots + r_n v_n.$$

For such a basis S, we define

$$\|x\|_\infty := \max_{1 \leq i \leq n} |r_i|.$$

Proposition 2.3. *Given a finite-dimensional vector space V and a basis $S = \{v_1, \cdots, v_n\}$ of V, the following hold:*

1. $(V, \|\cdot\|_\infty)$ *is Banach.*
2. *The set* $Q_M = \{x \in V :\ \|x\|_\infty \leq M\}$ *is compact.*
3. $K \subseteq (V, \|\cdot\|_\infty)$ *is compact if and only if K is closed and bounded.*

Proof. 1. Let x_k be a Cauchy sequence in V. Then $x_k = r_{k1}v_1 + \cdots + r_{kn}v_n$. For each i, the sequence $(r_{ki})_k$ is Cauchy in $\mathbb{R}$ since

$$|r_{ki} - r_{\ell i}| \leq \|x_k - x_\ell\|_\infty .$$

Hence, it converges to some r_i. Take the vector $x = r_1v_1 + \cdots + r_nv_n$. We get

$$\begin{aligned}\|x_k - x\|_\infty &= \|(r_{k1} - r_1)v_1 + \cdots + (r_{kn} - r_n)v_n\|_\infty \\ &\leq |r_{k1} - r_1| \, \|v_1\|_\infty + \cdots + |r_{kn} - r_n| \, \|v_n\|_\infty \to 0,\end{aligned}$$

concluding that x_k converges to x, and so $(V, \|\cdot\|_\infty)$ is Banach.

2. Take a sequence $x_k = r_{k1}v_1 + \cdots + r_{kn}v_n$ such that $\|x_k\|_\infty \leq M$. Then, for fixed i, $|r_{ki}| \leq M$ for every k; i.e., the sequences $(r_{ki})_k$ are bounded in $\mathbb{R}$. Applying Bolzano–Weierstrass n times, we construct a subsequence $r_{k_\ell i}$ that converges for every i. Let $r_i = \lim_{\ell\to\infty} r_{k_\ell i}$. Then, as in the proof of 1., x_{k_ℓ} converges to x, concluding that $\{x \in V \colon \|x\|_\infty \leq M\}$ is sequentially compact and hence compact.

3. ($\Longrightarrow$) is always true. ($\Longleftarrow$) Since K is bounded, there exists M such that

$$K \subseteq \{x \in V \colon \|x\|_\infty \leq M\} .$$

Since K is a closed subset of a compact set, we obtain that K is compact.

□

Theorem 2.1. *All norms on a finite-dimensional vector space V are equivalent.*

Remark 2.2. Note that the converse is not true in general. Take $P([a, b])$ as a subspace of $C([a, b])$ with the sup norm.

Proof. Take $S = \{v_1, \cdots, v_n\}$, a basis of V, and define

$$\|x\|_\infty = \max_{1\leq i\leq N} |r_i|.$$

Let $\|\cdot\|_a$ be a norm on V. It is enough to show that $\|\cdot\|_a$ is equivalent to $\|\cdot\|_\infty$.

We have

$$\begin{aligned}\|x\|_a &= \|r_1v_1 + \cdots + r_nv_n\|_a \\ &\leq |r_1| \, \|v_1\|_a + \cdots + |r_n| \, \|v_n\|_a \\ &\leq (\|v_1\|_a + \cdots + \|v_n\|_a) \, \|x\|_\infty .\end{aligned}$$

Letting $C_1 = \|v_1\|_a + \cdots + \|v_n\|_a$, we get

$$\|x\|_a \leq C_1\|x\|_\infty.$$

From the above inequality, the map $\varphi \colon (V, \|\cdot\|_\infty) \to \mathbb{R}$ given by $\varphi(x) = \|x\|_a$ is continuous. In fact, if $x_n \to x_0$ in $(V, \|\cdot\|_\infty)$, then

$$|\varphi(x_n) - \varphi(x_0)| = \big| \, \|x_n\| - \|x_0\| \, \big| \leq \|x_n - x_0\|_a \leq C_1 \, \|x_n - x_0\|_\infty \to 0.$$

Now, let $K = \{x \in V \colon \|x\|_\infty = 1\}$. Since K is compact (closed and bounded) in $(V, \|\cdot\|_\infty)$, $\varphi(K)$ is compact in $\mathbb{R}$. Thus, $m = \inf \varphi(K)$ is attained at some $x_0 \in K$. Notice that $\|x_0\|_\infty = 1$, so $x_0 \neq 0$, and hence

$$m = \|x_0\|_a > 0.$$

Let $x \in V$ with $x \neq 0$. Then $\dfrac{x}{\|x\|_\infty} \in K$, and so

$$\varphi\left(\frac{x}{\|x\|_\infty}\right) = \left\|\frac{x}{\|x\|_\infty}\right\|_a \geq m.$$

We hence obtain

$$\|x\|_a \geq m \|x\|_\infty .$$

□

Corollary 2.1. *Since all norms on a finite-dimensional vector space are equivalent, the following properties are independent of the choice of norm:*

- *Every finite-dimensional subspace of a normed space is complete (and hence closed).*
- *Compactness in finite-dimensional spaces is equivalent to being closed and bounded.*
- *Every bounded sequence in a finite-dimensional space has a convergent subsequence.*

We show next that the characterization of compact sets as closed and bounded holds only in finite-dimensional spaces. We first need the following lemma.

Lemma 2.1 (Riesz Lemma)**.** *Given a normed space $(V, \|\cdot\|)$ and a proper closed subspace $\mathcal{M}$ of V, for every $0 < \alpha < 1$ there exists $x \in V$ s.t. $\|x\| = 1$ and*

$$d(x, \mathcal{M}) \geq \alpha.$$

Remark 2.3. In the particular case where $\mathcal{M}$ is finite-dimensional, the distance $d(x, \mathcal{M})$ is attained for every $x \in V$. Therefore, in that case, there exists $x \in V$ such that $\|x\| = 1$ and $d(x, \mathcal{M}) = 1$.

Proof. Take $x_0 \notin \mathcal{M}$. Since $\mathcal{M}$ is closed, $d(x_0, \mathcal{M}) > 0$. Fix $0 < \alpha < 1$. There exists $m_0 \in \mathcal{M}$ s.t.

$$d(x_0, \mathcal{M}) \leq \|x_0 - m_0\| < \frac{d(x_0, \mathcal{M})}{\alpha}.$$

Define $x = \frac{x_0 - m_0}{\|x_0 - m_0\|}$. Note that $\|x\| = 1$, and for every $m \in \mathcal{M}$,

$$\|x - m\| = \left\|\frac{x_0 - m_0}{\|x_0 - m_0\|} - m\right\| = \frac{\|x_0 - (m_0 + m\|x_0 - m_0\|)\|}{\|x_0 - m_0\|}.$$

But $m_0 + m\|x_0 - m_0\| \in \mathcal{M}$, so

$$\|x - m\| \geq \frac{d(x_0, M)}{\|x_0 - m_0\|} > \alpha.$$

Therefore,

$$d(x, \mathcal{M}) = \inf_{m \in \mathcal{M}} \|x - m\| \geq \alpha.$$

□

Theorem 2.2. *Let $(V, \|\cdot\|)$ be a normed space. The unit sphere in V is compact if and only if V is finite-dimensional.*

Proof. ($\Longleftarrow$) Always true.

($\Longrightarrow$) Assume that V is infinite-dimensional. Then we can construct

$$E_1 = \operatorname{Span}\{v_1\}, \quad E_2 = \operatorname{Span}\{v_1, v_2\}, \quad \cdots,$$

where the E_i are finite-dimensional and $\{v_i\}_i$ is linearly independent. Then

$$E_1 \subsetneq E_2 \subsetneq E_3 \subsetneq \cdots,$$

and, being finite-dimensional, the E_i are closed.

Fix some $u_1 \in E_1$ with $\|u_1\| = 1$. Having constructed $u_n \in E_n$ with $\|u_n\| = 1$, by Riesz's Lemma there exists $u_{n+1} \in E_{n+1}$ such that $\|u_{n+1}\| = 1$ and

$$d(u_{n+1}, E_n) \geq 1/2.$$

We obtain as sequence $(u_n)_n$ on the unit sphere such that for $m \neq n$ we have

$$\|u_m - u_n\| \geq 1/2.$$

Therefore, (u_n) has no Cauchy (and hence no convergent) subsequence. Thus, the unit sphere is not sequentially compact and hence not compact. □

2.3 $\ell^p(\mathbb{R})$ Spaces

Lemma 2.2. *Given $1 < p, q < \infty$ such that $\frac{1}{p} + \frac{1}{q} = 1$, assume that $x, y \geq 0$ and $xy = 1$. Then*

$$\frac{x^p}{p} + \frac{y^q}{q} \geq 1,$$

with equality if and only if $x = y = 1$.

Proof. Let $f(x, y) = \frac{x^p}{p} + \frac{y^q}{q}$. We want to minimize $f(x, y)$ subject to the constraint $g(x, y) = xy = 1$. Since, on the closed constraint $xy = 1$, $f(x, 1/x) \to \infty$ as $x \to 0^+$ or $x \to \infty$, f attains a global minimum on $g(x, y) = 1$.

Using Lagrange multipliers, we solve

$$\begin{cases} \nabla f &= \lambda \nabla g, \\ xy &= 1, \end{cases} \implies \begin{cases} x^{p-1} &= \lambda y, \\ y^{q-1} &= \lambda x, \\ xy &= 1. \end{cases}$$

Then we get

$$x^p = \lambda xy = \lambda, \qquad y^q = \lambda xy = \lambda,$$

so $x = \lambda^{1/p}$ and $y = \lambda^{1/q}$. Hence,

$$1 = xy = \lambda^{\frac{1}{p} + \frac{1}{q}} = \lambda.$$

Therefore, $x = y = 1$, and the absolute minimum of f on $g(x, y) = 1$ is $f(1, 1) = 1$. □

Corollary 2.2 (Young inequality)**.** *Given $1 < p, q < \infty$ such that $\frac{1}{p} + \frac{1}{q} = 1$, then*

$$\frac{|x|^p}{p} + \frac{|y|^q}{q} \geq |xy|,$$

with equality if and only if $|x|^p = |y|^q$.

Proof. Without loss of generality, assume $x \neq 0$ and $y \neq 0$. Take

$$X = \frac{|x|}{|xy|^{1/p}} \qquad \text{and} \qquad Y = \frac{|y|}{|xy|^{1/q}}.$$

Notice that

$$XY = \frac{|xy|}{|xy|^{1/p+1/q}} = 1.$$

Applying the lemma, we get

$$1 \leq \frac{X^p}{p} + \frac{Y^q}{q} = \frac{|x|^p}{|xy|p} + \frac{|y|^q}{|xy|q}.$$

Hence,

$$\frac{|x|^p}{p} + \frac{|y|^q}{q} \geq |xy|,$$

with equality if and only if $X = Y = 1$. This is equivalent to

$$|x|^p = |xy| = |y|^q.$$

□

Corollary 2.3 (Hôlder inequality)**.** *Given $1 < p, q < \infty$ such that $\frac{1}{p} + \frac{1}{q} = 1$, and sequences x_n and y_n, then*

$$\sum_{i=1}^{n} |x_i y_i| \leq \left(\sum_{i=1}^{n} |x_i|^p \right)^{1/p} \left(\sum_{i=1}^{n} |y_i|^q \right)^{1/q}.$$

Letting $n \to \infty$ (avoiding $0 \times \infty$), we obtain

$$\sum_{n=1}^{\infty} |x_n y_n| \leq \left(\sum_{n=1}^{\infty} |x_n|^p \right)^{1/p} \left(\sum_{n=1}^{\infty} |y_n|^q \right)^{1/q}.$$

Proof. Without loss of generality, assume that not all x_i and not all y_i are zero. Take

$$X_i = \frac{|x_i|}{\left(\sum_{k=1}^{n} |x_k|^p\right)^{1/p}} \qquad \text{and} \qquad Y_i = \frac{|y_i|}{\left(\sum_{k=1}^{n} |y_k|^q\right)^{1/q}}.$$

From Young's inequality, we have

$$\begin{aligned} X_i Y_i &= \frac{|x_i||y_i|}{\left(\sum_{k=1}^{n} |x_k|^p\right)^{1/p} \left(\sum_{k=1}^{n} |y_k|^q\right)^{1/q}} \\ &\leq \frac{X_i^p}{p} + \frac{Y_i^q}{q} \\ &= \frac{|x_i|^p}{\left(\sum_{k=1}^{n} |x_k|^p\right) p} + \frac{|y_i|^q}{\left(\sum_{k=1}^{n} |y_k|^q\right) q}. \end{aligned}$$

Therefore,

$$\sum_{i=1}^{n} X_i Y_i \leq \frac{1}{p} + \frac{1}{q} = 1.$$

Substituting the definitions of X_i and Y_i gives the desired inequality. □

Corollary 2.4 (Minkowski inequality)**.** *Given sequences x_n and y_n, and $p \geq 1$, then*

$$\left(\sum_{i=1}^{n} |x_i + y_i|^p\right)^{1/p} \leq \left(\sum_{i=1}^{n} |x_i|^p\right)^{1/p} + \left(\sum_{i=1}^{n} |y_i|^p\right)^{1/p}.$$

Letting $n \to \infty$, we obtain

$$\left(\sum_{n=1}^{\infty} |x_n + y_n|^p\right)^{1/p} \leq \left(\sum_{n=1}^{\infty} |x_n|^p\right)^{1/p} + \left(\sum_{n=1}^{\infty} |y_n|^p\right)^{1/p}.$$

Proof. The case $p = 1$ follows from the triangle inequality.

Take $p > 1$ and let q be such that $\frac{1}{p} + \frac{1}{q} = 1$. Then

$$\begin{aligned}
\sum_{i=1}^{n} |x_i + y_i|^p &= \sum_{i=1}^{n} |x_i + y_i| \cdot |x_i + y_i|^{p-1} \\
&\leq \sum_{i=1}^{n} \left(|x_i| + |y_i|\right) |x_i + y_i|^{p-1} \\
&= \sum_{i=1}^{n} |x_i||x_i + y_i|^{p-1} + \sum_{i=1}^{n} |y_i||x_i + y_i|^{p-1} \\
&\leq \left(\sum_{i=1}^{n} |x_i|^p\right)^{1/p} \left(\sum_{i=1}^{n} |x_i + y_i|^{(p-1)q}\right)^{1/q} + \left(\sum_{i=1}^{n} |y_i|^p\right)^{1/p} \left(\sum_{i=1}^{n} |x_i + y_i|^{(p-1)q}\right)^{1/q}.
\end{aligned}$$

We have

$$\frac{1}{p} + \frac{1}{q} = 1 \implies p + q = pq \implies p = q(p-1).$$

Therefore,

$$\sum_{i=1}^{n} |x_i + y_i|^p \leq \left(\sum_{i=1}^{n} |x_i + y_i|^p\right)^{1/q} \left(\left(\sum_{i=1}^{n} |x_i|^p\right)^{1/p} + \left(\sum_{i=1}^{n} |y_i|^p\right)^{1/p}\right).$$

If $x_i + y_i = 0$ for every i, the inequality holds trivially. Otherwise, dividing both sides of the above inequality by $\left(\sum_{i=1}^{n} |x_i + y_i|^p\right)^{1/q}$ gives the desired Minkowski inequality. □

Definition 2.5. Let $1 \leq p < \infty$. We define the space

$$\ell^p(\mathbb{R}) = \left\{ x = (x_1, x_2, \ldots) \colon \sum_{n=1}^{\infty} |x_n|^p < \infty \right\},$$

with the norm

$$\|x\|_p = \left(\sum_{n=1}^{\infty} |x_n|^p\right)^{1/p}.$$

This is indeed a norm by Minkowski's inequality.

As in the case $p = 1$, we can prove the following.

Proposition 2.4. *Given the $\ell^p(\mathbb{R})$ spaces defined above for $1 \leq p < \infty$, we have the following:*

- *$\ell^p(\mathbb{R})$ is Banach and separable.*
- *$K \subseteq \ell^p(\mathbb{R})$ is compact if and only if K is closed and bounded and*

$$\lim_{N\to\infty} \sup_{x\in K} \sum_{n=N+1}^{\infty} |x_n|^p = 0.$$

- *If $1 \leq r < s < \infty$, then*

$$\ell^r(\mathbb{R}) \subsetneq \ell^s(\mathbb{R}).$$

Proof. The first two properties follow very similarly to the case $p = 1$.

Notice that if $r < s$ and $x = (x_1, x_2, \ldots) \in \ell^r(\mathbb{R})$, then $\lim_{n\to\infty} x_n = 0$. Therefore, there exists N such that for $n \geq N$, $|x_n| < 1$, which implies that

$$|x_n|^s \leq |x_n|^r.$$

Hence, by the comparison test, $x \in \ell^s(\mathbb{R})$. Moreover, notice that

$$x_n = \frac{1}{n^{1/r}} \in \ell^s(\mathbb{R}) \setminus \ell^r(\mathbb{R}).$$

□

Remark 2.4. Recall that $\ell^\infty(\mathbb{R})$ is complete but is not separable. Also, if $1 \leq p < \infty$, then $\sum_{n=1}^{\infty} |x_n|^p < \infty$ implies that $x_n \to 0$. Hence, for such p,

$$\ell^p(\mathbb{R}) \subsetneq c_0(\mathbb{R}) \subsetneq \ell^\infty(\mathbb{R}).$$

The first inclusion is strict since

$$x_n = \frac{1}{n^{1/p}} \in c_0(\mathbb{R}) \setminus \ell^p(\mathbb{R}),$$

and the second inclusion is strict since $\mathbf{1} = (1, 1, \cdots) \in \ell^\infty(\mathbb{R}) \setminus c_0(\mathbb{R})$.

2.4 Bounded Linear Transformations

Definition 2.6. Given vector spaces V, W over $\mathbb{F}$, a linear transformation $T : V \mapsto W$ satisfies

$$T(x + \alpha y) = Tx + \alpha Ty \qquad \forall x, y \in V,\ \alpha \in \mathbb{F}.$$

We denote by $\mathcal{L}(V, W)$ the space of linear transformations from V to W.

Remark 2.5. $\mathcal{L}(V,W)$ is a vector space over $\mathbb{F}$. Moreover, when V and W are finite dimensional, $\mathcal{L}(V,W)$ is finite dimensional and

$$\dim \mathcal{L}(V,W) = \dim(V) \cdot \dim(W).$$

Definition 2.7. Take $(V, \|\cdot\|_V)$ and $(W, \|\cdot\|_W)$, and let $T \in \mathcal{L}(V,W)$. We say that T is bounded if and only if there exists $M > 0$ such that

$$\|Tx\|_W \leq M \|x\|_V .$$

Equivalently, $\exists M$ such that

$$\|Tx\|_W \leq M, \quad \forall x \text{ such that } \|x\|_V = 1.$$

Proposition 2.5. *Let $(V, \|\cdot\|_V)$ and $(W, \|\cdot\|_W)$ be normed spaces and $T \in \mathcal{L}(V,W)$. Then*

$$T \text{ bounded} \iff T \text{ is continuous} \iff T \text{ is continuous at } 0.$$

Proof.

- Assume T is bounded. Then there exists $M > 0$ such that $\|Tx\| \leq M \|x\|$. Hence,

  $$\|Tx - Ty\| = \|T(x-y)\| \leq M \|x-y\| .$$

  Thus, T is Lipschitz, so it is continuous.

- Assume T is continuous. Then, trivially, it is continuous at 0.

- If T is continuous at 0, then there exists $\delta > 0$ such that $\|Tx - T0\|_W < 1$ whenever $\|x - 0\|_V < \delta$. This gives

  $$\|Tx\|_W \leq 1$$

  for $\|x\|_V < \delta$. Now take $x \neq 0$. Then

  $$\left\| T\left(\frac{\delta}{2} \frac{x}{\|x\|} \right) \right\| \leq 1.$$

  Thus, we get, for every x,

  $$\|Tx\| \leq \frac{2}{\delta} \|x\| .$$

□

Remark 2.6. For a finite dimensional space $(V, \|\cdot\|_V)$, every linear transformation T is bounded (and hence continuous). In fact, let $S = \{v_1, \cdots, v_n\}$ be a basis of V, and take $x = r_1 v_1 + \cdots + r_n v_n \in V$. Then

$$\|Tx\|_W \leq |r_1| \|Tv_1\|_W + \cdots + |r_n| \|Tv_n\|_W \leq (\|Tv_1\|_W + \cdots + \|Tv_n\|_W) \|x\|_\infty ,$$

where $\|x\|_\infty = \max |r_i|$. Since all norms on V are equivalent, we then have

$$\|Tx\|_W \leq M \|x\|_V .$$

2.4.1 The Operator Norm

Definition 2.8. Given normed spaces $(V, \|\cdot\|)$ and $(W, \|\cdot\|)$, we define $B(V,W)$ to be the space of bounded linear transformations. This is, in fact, a subspace of $\mathcal{L}(V,W)$. We define the corresponding operator norm by

$$\|T\|_{B(V,W)} = \sup_{\substack{x\in V \\ \|x\|_V=1}} \|Tx\|_W = \sup_{\substack{x\in V \\ x\neq 0}} \frac{\|Tx\|_W}{\|x\|_V}.$$

Remark 2.7. Notice that for $x \in V$, $x \neq 0$, we have

$$\left\| T\frac{x}{\|x\|_V} \right\|_W \leq \|T\|_{B(V,W)}.$$

Hence, for every $x \in V$,

$$\|Tx\|_W \leq \|T\|_{B(V,W)} \cdot \|x\|_V.$$

Also, for every $x, y \in V$,

$$\|Tx - Ty\|_W \leq \|T\|_{B(V,W)} \cdot \|x - y\|_V.$$

Hence, $\|T\|_{B(V,W)}$ represents the optimal Lipschitz constant corresponding to T.

Proposition 2.6. *If W is Banach, then $B(V,W)$ is Banach.*

Proof. Let T_n be a Cauchy sequence in $B(V,W)$. Notice that

$$\|T_n x - T_m x\| \leq \|T_n - T_m\| \cdot \|x\|.$$

Since T_n is Cauchy in $B(V,W)$, $T_n x$ is Cauchy in W, which is Banach, and hence it converges. We then define

$$T\colon V \to W, \quad T(x) = \lim_{n\to\infty} T_n(x).$$

Notice that T is linear. In fact,

$$T(x + \alpha y) = \lim_{n\to\infty} T_n(x + \alpha y) = \lim_{n\to\infty} (T_n x + \alpha T_n y) = Tx + \alpha Ty.$$

We next show that T is bounded. In fact, since $\{T_n\}$ is Cauchy in $B(V,W)$, it is bounded, and so there exists $M > 0$ such that $\|T_n\| \leq M$ for all n. Now,

$$\|T_n x\| \leq \|T_n\| \cdot \|x\| \leq M \|x\|.$$

Letting $n \to \infty$, we get $\|Tx\| \leq M\|x\|$, so $T \in B(V,W)$.

It remains to prove that $T_n \to T$ in $B(V,W)$. Given $\varepsilon > 0$, since T_n is Cauchy, there exists N_1 such that

$$\|T_n - T_m\| < \varepsilon/2, \quad \forall n, m \geq N_1.$$

Now fix $x \in V$ such that $\|x\| = 1$. Since $T_n x \to Tx$ in W, there exists $N_2 > N_1$ such that

$$\|T_{N_2} x - Tx\| < \varepsilon/2.$$

Hence, for $n \geq N_1$,

$$\begin{aligned} \|T_n x - Tx\| &\leq \|T_n x - T_{N_2} x\| + \|T_{N_2} x - Tx\| \\ &\leq \|T_n - T_{N_2}\| \cdot \|x\| + \frac{\varepsilon}{2} < \varepsilon. \end{aligned}$$

We conclude that for all $n \geq N_1$ and all $x \in V$ with $\|x\| = 1$, we have

$$\|T_n x - Tx\| < \varepsilon.$$

Therefore,

$$\sup_{\substack{x \in V \\ \|x\|=1}} \|T_n x - Tx\| \leq \varepsilon, \quad \forall n \geq N_1.$$

Hence, $T_n \to T$ in $B(V, W)$. □

Example 2.3. Define

$$T\colon\ (C([0,1]), \|\cdot\|_\infty) \mapsto \mathbb{R}, \quad T(f) = \int_0^1 f(x)\,dx.$$

Show that $T \in B(C([0,1]), \mathbb{R})$ and find $\|T\|$.

Solution. Clearly, T is linear because integration is linear. Now,

$$|Tf| = \left| \int_0^1 f(x)\,dx \right| \leq \int_0^1 |f(x)|\,dx \leq \|f\|_\infty .$$

Thus, T is bounded and

$$\|T\| = \sup_{\|f\|_\infty = 1} |Tf| \leq 1.$$

On the other hand, $f_0 = 1 \in C[0,1]$, $\|f_0\|_\infty = 1$, and $|Tf_0| = 1$. Therefore, $\|T\| = 1$.

2.4.2 Dual Space

Definition 2.9. Let V be a normed space over $\mathbb{F} = \mathbb{R}$ or $\mathbb{C}$. We define the dual space $V^* = B(V, \mathbb{F})$, which is the space of all bounded linear maps from V to $\mathbb{F}$. We note that such maps are sometimes called bounded linear functionals. The corresponding operator norm is called the dual norm:

$$\|f\|_* = \sup_{\substack{x \in V \\ \|x\|=1}} |f(x)| = \sup_{\substack{x \in V \\ x \neq 0}} \frac{|f(x)|}{\|x\|}.$$

Notice that since $\mathbb{F}$ is Banach, the dual space V^* is always a Banach space.

Remark 2.8. If V is finite dimensional, then $V^* = B(V, \mathbb{F}) = \mathcal{L}(V, \mathbb{F})$, and

$$\dim V^* = \dim V < \infty.$$

Therefore, V^* and V are isomorphic.

Notation. Given $(V, \|\cdot\|_V)$ and $(W, \|\cdot\|_W)$, we write $V \cong W$ if and only if V and W are isometrically isomorphic, i.e., there exists an isomorphism $\varphi\colon V \to W$ that is also an isometry.

Theorem 2.3. *We have*

$$(\ell^1)^* \cong \ell^\infty.$$

Moreover, for $1 < p < \infty$,

$$(\ell^p)^* \cong \ell^q,$$

where q *is such that* $\dfrac{1}{p} + \dfrac{1}{q} = 1$.

Remark 2.9. We make a few observations.

- For $p = 2$, we get
$$(\ell^2)^* \cong \ell^2.$$
- One can show that if $V \cong W$, then $V^* \cong W^*$. This implies that, for $1 < p < \infty$,
$$(\ell^p)^{**} \cong (\ell^q)^* \cong \ell^p.$$
- The dual of ℓ^∞ is not ℓ^1, as we will show in the coming chapter, so the above property fails when $p = 1$.

To prove the theorem, we will use the following lemma.

Lemma 2.3. *Let e_i be the sequence whose i-th element is 1 and whose remaining elements are 0. If $1 \le p < \infty$, then for every $x = (x_1, x_2, \cdots) \in \ell^p$, we have*
$$x = \sum_{n=1}^{\infty} x_n e_n = \lim_{N\to\infty} \sum_{n=1}^{N} x_n e_n.$$

Proof. Take $x = (x_1, x_2, \cdots) \in \ell^p$. Notice that
$$\sum_{n=1}^{N} x_n e_n = (x_1, \cdots, x_N, 0, 0, \cdots).$$
Then
$$\left\| x - \sum_{n=1}^{N} x_n e_n \right\|_p = \|(0, 0, \cdots, 0, x_{N+1}, x_{N+2}, \cdots)\|_p = \left(\sum_{n=N+1}^{\infty} |x_n|^p \right)^{1/p} \to 0,$$
since it is the tail of a convergent series. □

Remark 2.10. Above statements is not true for $p = \infty$. For example, taking $x = \mathbf{1} = (1, 1, 1, \cdots)$. Then $\|x - \sum_{n=1}^{N} x_n e_n\|_\infty = 1$, which does not go to zero.

Proof of the duality of ℓ^1. Given $T \in (\ell^1)^*$, by the continuity of T and the above lemma, for every $x \in \ell^1$,
$$Tx = T\left(\sum_{n=1}^{\infty} x_n e_n \right) = \sum_{n=1}^{\infty} x_n T e_n.$$
Define the sequence $b = (Te_1, Te_2, \cdots)$. Notice that
$$|Te_i| \le \|T\|_* \|e_i\|_1 = \|T\|_*.$$
Thus, $b \in \ell^\infty$ and $\|b\|_\infty \le \|T\|_*$. On the other hand,
$$|Tx| = \left| \sum_{n=1}^{\infty} x_n T e_n \right| \le \|b\|_\infty \|x\|_1.$$
Therefore, $\|T\|_* \le \|b\|_\infty$, and hence
$$\|T\|_* = \|b\|_\infty.$$

Using all of the above, we deduce that the map $\psi : (\ell^1)^* \mapsto \ell^\infty$, defined by

$$\psi(T) = (Te_1, Te_2, \cdots),$$

is an isometry. It is also clearly linear. It remains to show that it is surjective.

In fact, for every $b \in \ell^\infty$, consider the map $T_b : \ell^1 \mapsto \mathbb{R}$ defined by

$$T_b(x) = \sum_{n=1}^{\infty} x_n b_n.$$

Clearly, the right-hand side converges, T_b is linear, and

$$|T_b(x)| \leq \|b\|_\infty \|x\|_1,$$

implying that $T_b \in (\ell^1)^*$. Notice that

$$\psi(T_b) = (T_b e_1, T_b e_2, \cdots) = b,$$

showing that ψ is surjective. □